

Maryland UAS Technical Design Report 2026

Maryland Unmanned Aerial Systems (MUAS)

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Abstract—This paper presents the competition strategy, system design, and testing approach for TUZA, the Maryland Unmanned Aerial Systems team’s fixed-wing autonomous aircraft developed for the 2026 SUAS Competition. The design prioritizes high-value mission objectives, particularly completion of ten 2-mile endurance laps, while balancing reliability, system complexity, weight, transportability, and available development resources. A Heewing T2 Cruza airframe was selected for its aerodynamic efficiency, internal volume, and modular assembly. TUZA integrates a 3DR Control Zero H7 flight controller, NVIDIA Jetson Orin Nano companion computer, multi-link communication architecture, SIYI A8 Mini camera, and a custom 3D-printed payload-release and landing skid assembly. During the survey mission, the aircraft records high-resolution imagery for ground-based OpenDroneMap processing while a custom-trained YOLO26 model detects and estimates the locations of mission targets. Confirmed coordinates are then used to generate a subsequent autonomous payload-delivery mission. System validation follows a staged testing strategy that progresses from component-level bench tests and SITL simulation to subsystem integration and full-flight testing. This approach is intended to reduce technical risk while demonstrating repeatable performance across endurance, mapping, target identification, communications, and payload-delivery operations.

Index Terms—Unmanned Aircraft Systems, Fixed-wing UAS, Obstacle Avoidance, ArduPilot, SUAS competition

I. COMPETITION STRATEGY

A. Strategic Vision

MUAS’s strategic vision for this year’s competition focused on maximizing as many points as possible while operating within the constraints of the team’s available manpower, experience, budget, and development timeline. Following the release of the Team Handbook in late October, the team analyzed each mission objective and its associated point value to determine where engineering effort would have the greatest impact. A decision matrix

was developed to evaluate each capability based on several factors, including point value, implementation complexity, technical risk, required development time, and overall feasibility. This structured approach allowed the team to prioritize features that offered the highest return on investment while minimizing the risk of introducing unnecessary system complexity.

B. Platform Selection and Scoring Trade-Offs

A fixed-wing platform was ultimately selected as the competition platform, as it would allow the team to earn the most points given its experience and available resources. Compared to a multirotor platform, a fixed-wing aircraft offers significantly greater flight time, providing a much higher likelihood of meeting the competition’s flight endurance objective. This would allow the team to complete the ten 2-mile laps needed to earn the full 200 points allocated for flight endurance.

The team recognized that this decision introduced trade-offs. While a multirotor aircraft would simplify precision payload delivery and low-speed maneuvering, its reduced endurance would significantly limit the ability to complete the targeted 10 laps for the endurance task. Additionally, the superior aerodynamic efficiency of the fixed-wing platform reduced the aircraft’s overall energy requirements. Since the team elected to power the UAS exclusively with sub-100 Wh battery packs, the improved efficiency enabled the required endurance to be achieved with fewer battery packs than on an equivalent multirotor platform. This reduced the overall battery weight, allowing the aircraft to target the maximum 50 points for all-up weight while simultaneously earning the full 75 points available for the Easy to Transport Batteries objective. Furthermore, the selected airframe incor-

TABLE I
PLATFORM DECISION MATRIX (WEIGHTED SCORING)

Criterion	Max Pts	Weight (%)	Quadcopter		Kraken		Fixed-Wing		Other Hybrid	
			Feas.	Score	Feas.	Score	Feas.	Score	Feas.	Score
Flight Endurance (≥ 1 successful takeoff (above 150 ft AGL) and landing, manual or autonomous)	250	25.0	10	20	9	16.2	9	16.2	9	14.6
All Takeoffs/Landings Conducted Autonomously	20	3.0	8	24	8	16.8	4	6	6	16.2
Flight Endurance (Number of Laps, up to 10 laps @ approx. 2 mi each)	200	20.0	3	9	5	40	10	160	8	102.4
Efficient Operators (2–4 operators)	200	20.0	10	200	10	200	10	200	8	128
Design for Rapid Response / UAS All-Up Weight (≤ 35 lbs)	50	5.0	6	21	8	28	9	45	6	15
UAS Volume ($18 \times 14 \times 8$ in. personal item / $22 \times 14 \times 8$ in. carry-on / $27 \times 21 \times 14$ in. checked)	50	5.0	8	22.5	6	7	7	17.5	6	5
UAS Volume (Tool-less Assembly)	25	2.5	10	20	8	6.7	7	18	6	7.5
Easy to Transport Batteries (≤ 100 Wh)	75	7.5	8	42	6	36.7	8	48	8	36
Risk Mapping	150	15.0	7	84	7	52.5	5	37.5	7	73.5
Search, Detect, and Deliver (2 each)	200	20.0								
Object Survives (within vicinity of search boundary)	20	2.0	10	40	10	40	10	32	10	40
Object Lands within 50 ft of target	50	5.0	8	81	8	72	8	24	8	72
Object Delivered to Correct Target (water bottle to tent, beacon)	30	3.0	9	43.2	8	33.6	4	12	7	33.6
Weighted Total (out of 1000)										
										Quadcopter 606.7
										Kraken 553.35
										Fixed-Wing 616.2
										Other Hybrid 543.6

porates a modular, tool-less assembly, enabling the team to pursue the additional 25 points awarded for tool-less assembly.

Although the UAS Volume objective awards up to 50 points, the larger wingspan of the selected fixed-wing platform exceeded the dimensional limits of the Personal Item category ($18 \times 14 \times 8$ in.). Rather than compromising aerodynamic performance and structural integrity for an additional 25–40 points, the team accepted a lower score in the UAS Volume objective, recognizing that the additional points available for a smaller aircraft did not justify the associated reduction in overall mission capability and reliability.

C. Software Strategy

The software strategy focuses on coordinating mapping, target identification, payload delivery, and endurance objectives without adding unnecessary complexity to the aircraft. TUZA will complete the ten endurance laps and required mapping passes while transmitting imagery, GPS position, aircraft orientation, and timing information to the ground control station (GCS). This provides real-time situational awareness and allows the team to verify image coverage and camera operation during the mission.

At the same time, the SIYI A8 Mini will record higher-resolution 4K footage to its onboard SD card. Because the onboard recording provides substantially better image quality than the live stream, the SD card footage will serve as the primary source for map generation. At the same time, a custom-trained YOLO26 model analyzes the live video

feed for objects of interest. Confirmed detections and their estimated coordinates are recorded for use in generating the autonomous payload-delivery mission.

After completing the endurance and mapping tasks, TUZA will return and land so that the SD card can be retrieved. Performing photogrammetry processing after landing avoids placing unnecessary computational and power demands on the aircraft during flight. TUZA will then be relaunched to execute the generated autonomous mission, navigate to the confirmed target locations, and deliver the corresponding payloads.

II. DESIGN STRATEGY



Fig. 1. TUZA, the MUAS fixed-wing autonomous aircraft.

A. Airframe and Propulsion

As a newly reestablished team with limited experience designing an airframe from the ground up, the team opted to utilize an off-the-shelf airframe as the foundation of the UAS. This architectural decision reduced development time and technical

risk while allowing the team to concentrate engineering effort on higher-value competition objectives, including autonomy, software integration, and mission-specific subsystem development.

Both custom and off-the-shelf airframes were considered during the conceptual design phase. Although a custom airframe would have provided greater flexibility in aerodynamic geometry, layout, and component selection, it would also have required substantial effort in aerodynamic design, structural analysis, manufacturing, and flight validation. Given the team's size, available manufacturing resources, budget, and development schedule, the potential benefits of a custom platform did not justify the associated cost and risk. A proven off-the-shelf airframe was therefore determined to provide the most practical path toward a reliable, fully integrated competition UAS.

Four commercially available fixed-wing platforms – the Heewing T2 Cruza, AtomRC Dolphin Pro, AtomRC Dolphin V1.1, and TBS Mojito – were evaluated through a comparative trade study. Each candidate was assessed based on projected endurance, empty weight, wing loading, cruise power, battery compatibility, available internal volume, payload capacity, cost, and ease of assembly. These criteria were selected to evaluate both aircraft performance and the ability to integrate the required avionics, batteries, wiring, onboard electronics, and mission-specific hardware.

Manufacturer specifications, publicly available build logs, and flight demonstration videos were reviewed to establish baseline performance data, including cruise speed and current draw, and to identify representative component configurations for estimating base aircraft weight. A preliminary power analysis was conducted for each candidate using a proposed 6S battery (22.2 V) with an initial proposed capacity of 10 Ah at the estimated competition-ready weight. Because the cruise-current values were generally based on lighter reference configurations, current draw was adjusted in proportion to the estimated increase in aircraft weight as a first-order approximation. The adjusted cruise current was then used to estimate the endurance and range of each airframe under the team's expected competition configuration, assuming a usable battery fraction of 0.8 to ensure safety

and reliability.

Based on this evaluation, the Heewing T2 Cruza was selected as the competition airframe. It provided the strongest combination of projected endurance, internal packaging volume, tool-less assembly, and sufficient wing area and payload capacity to handle the additional batteries, mission payload, and on-board electronics while maintaining an acceptable wing loading.

TABLE II
AIRFRAME TRADE STUDY SUMMARY ASSUMING A 6S, 10 AH BATTERY CONFIGURATION

Parameter	Dolphin Pro	T2 Cruza	Mojito
Cruise Speed	22 m/s	19 m/s	33 m/s
Endurance	40 min	51 min	26 min
Range	53 km	58 km	52 km

The T2 Cruza is primarily constructed from EPP foam, with carbon-fiber rods reinforcing the wings and a removable carbon-fiber tail boom that supports rapid, tool-less assembly. The propulsion system utilizes two Heewing T2 FX-2216-850KV motors paired with APC 8x6E propellers and Heewing FX-25A ESCs [2].

B. Component Selection

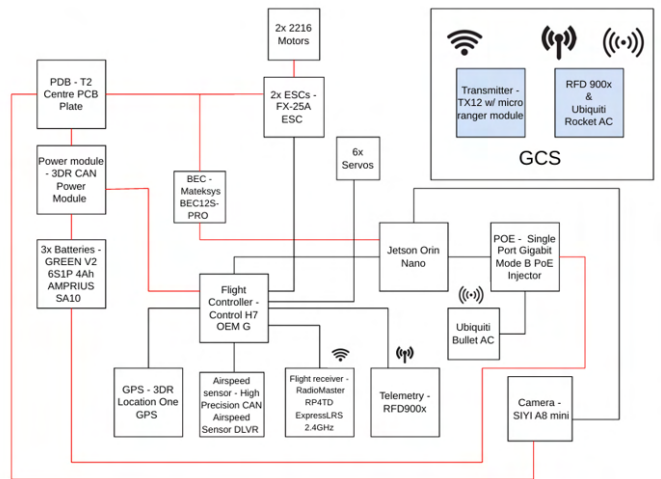


Fig. 2. TUZA power, avionics, and communication architecture block diagram.

1) *Autopilot and Peripherals:* TUZA's autopilot is handled by the 3DR Control Zero H7 OEM Rev G flight controller paired with a 3DR reference-design

carrier board running ArduPlane. The Control Zero H7 was selected for its compact 2.42 in $\times$ 1.28 in footprint, lighter weight, and connectivity interface, which was better suited to TUZA's limited space and subsystem architecture than a Cube Orange+ with a mini carrier board. It includes three IMUs, a barometer, and a magnetometer to provide attitude, altitude, and heading measurements [3], [4].

2) *Communication and Navigation*: The communication and navigation architecture supports command and control, telemetry, GNSS positioning, image and video transmission, and Remote ID broadcasting. For Remote ID, the team selected the FT EZ ID because of its straightforward setup process, an average current draw of 10 mA, and its ability to transmit up to distances of 1500 ft.

To support telemetry and communication with the GCS, TUZA uses a multi-link communication architecture. Two RFD900x (900 MHz) modules provide a long-range telemetry link for transmitting flight data and mission commands between the aircraft and the GCS. For higher-bandwidth communication such as image and video transmission, a Ubiquiti Bullet AC (5 GHz) installed on the aircraft communicates with a Ubiquiti R5AC-LITE Rocket AC (5 GHz) at the GCS.

A 3DR Location One module provides GNSS positioning and heading data for the UAS through its CAN interface, allowing straightforward integration with existing hardware. For manual RC control by the safety pilot, an RP4TD ExpressLRS RC receiver (2.4 GHz) is installed on the UAS and communicates with a RadioMaster TX12 MKII radio controller. Since the team's TX12 is a CC2500 version that lacks native ELRS support, an external RadioMaster Ranger Micro ELRS module was installed, allowing the controller to communicate with the receiver while also extending transmitter range.

One of the primary communication-system design challenges was selecting antennas for the 5 GHz video link. The airMAX Rocket AC Lite specifications show a trade-off between data rate and receiver sensitivity: higher data rates correspond to lower sensitivity, while lower data rates correspond to higher sensitivity. To offset this, maximizing antenna gain at both the ground station and the UAS is critical [5].

For the Ubiquiti Bullet AC on the UAS, the

TABLE III
R5AC-LITE OUTPUT POWER (27 dBm) – TX/RX
SPECIFICATIONS

Modulation	Data Rate	Avg. TX	Sensitivity	Tol.
airMAX ac	1x BPSK (1/2)	27 dBm	−96 dBm	±2 dB
	2x QPSK (1/2)	27 dBm	−95 dBm	±2 dB
	2x QPSK (3/4)	27 dBm	−92 dBm	±2 dB
	4x 16QAM (1/2)	27 dBm	−90 dBm	±2 dB
	4x 16QAM (3/4)	27 dBm	−86 dBm	±2 dB
	6x 64QAM (2/3)	27 dBm	−83 dBm	±2 dB
	6x 64QAM (3/4)	26 dBm	−77 dBm	±2 dB
	6x 64QAM (5/6)	25 dBm	−74 dBm	±2 dB
	8x 256QAM (3/4)	23 dBm	−69 dBm	±2 dB
	8x 256QAM (5/6)	22 dBm	−65 dBm	±2 dB

team selected the Skyzone 5.8 GHz FPV mushroom antenna. It offers an average gain of 3.71 dBi, well above the typical 2 dBi baseline of competing options. An omnidirectional pattern was chosen to maximize spatial coverage toward the ground station during typical flight maneuvers, with special consideration for the antenna null points directly above and below the antenna axis during steep tilts or overhead passes.

Antenna polarization was another key consideration. Linear (horizontal or vertical) polarization suffers from polarization mismatch loss during dynamic pitch, roll, or bank maneuvers. Circular polarization (CP) mitigates these orientation losses and significantly reduces multipath interference. When a circularly polarized wave (e.g., RHCP) bounces off an obstacle, its polarization reverses (to LHCP); because the receiving antenna is designed for the original polarization, it naturally rejects the reflected signal, preventing it from interfering with the direct transmission [6].

The Ubiquiti airMAX Rocket AC employs a directional patch antenna paired with an antenna tracker to boost directivity and overall gain. Directional antennas require a compromise between gain and beamwidth (lobe angle): a narrow beamwidth yields higher gain but demands extreme tracking accuracy to prevent signal loss, whereas a wider beamwidth offers a larger target area at the expense of gain. The team selected the ZRC 5.8 GHz Circular Polarized High Gain Flat Panel Antenna, which delivers a strong 14 dBi gain while maintaining a forgiving 40° beamwidth.

Finally, the team evaluated operating-frequency options (2.4 GHz vs. 5.8 GHz). While 2.4 GHz offers better signal propagation and obstacle penetration, the 5.8 GHz band provides wider available channel bandwidth, enabling significantly higher data rates. Using a free-space-path-loss (FSPL) calculator, the team verified that 5.8 GHz provides sufficient link margin to meet the target range requirements using the formula below [7]:

$$\text{FSPL} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}\left(\frac{4\pi}{c}\right) - G_{Tx} - G_{Rx} \quad (1)$$

3) *Edge Computer*: An NVIDIA Jetson Orin Nano is the onboard companion computer, supporting the team's computer vision pipeline for target detection, payload-delivery operations, and photogrammetry. The Jetson was chosen for its real-time AI processing capability, CUDA architecture, and TensorRT acceleration. The team selected the Seeed Studio A603 carrier board because its compact 87×52 mm footprint is substantially smaller than the NVIDIA Jetson Orin Nano Developer Kit's 103×90.5 mm footprint, making it better suited to TUZA's limited fuselage space [8].

4) *Power*: Power for the UAS is provided by three Upgrade Energy GREEN V2 6S1P 4 Ah lithium-ion batteries [9] connected in parallel, providing a combined capacity of 12 Ah. Each battery is rated at 86 Wh, below the FAA's 100 Wh limit for lithium-ion batteries carried by airline passengers [10].

Battery selection for TUZA balanced weight, form factor, energy density, and discharge capability. The compact $7.0 \times 3.8 \times 5.7$ cm packs can be distributed throughout the fuselage to support center-of-gravity management. Together, the three batteries provide 18 A continuously, 33 A for two minutes, and 48 A for ten seconds, exceeding the expected current demand during cruise, turns, and takeoff.

TUZA's onboard avionics and peripherals operate across multiple voltage levels, including the unregulated 6S battery bus, a regulated 12 V mission-system rail, and a regulated 5 V avionics rail. Power is initially distributed through the included Heewing T2 Center PCB Plate, which serves as the primary power distribution board (PDB) and supplies the

6S bus to the ESCs, motors, and the PoE supply for the onboard Ubiquiti Bullet AC. Although the board included a regulated 12 V output that could supply the 12 V rail needed for the Jetson, SIYI A8 Mini gimbal camera, and FT EZ ID Remote ID module, limited documentation regarding its continuous output current capability led the team to elect not to use these outputs for mission-critical electronics.

Instead, a dedicated Mateksys BEC 12S-Pro was integrated into the power system to provide the regulated 12 V rail. The step-down converter is rated for a continuous output current of 5 A and a peak output current of 9 A, providing sufficient margin for transient loads.

A 3DR CAN power module provides the regulated 5 V rail for the flight controller while also measuring battery voltage and current. These measurements enable real-time power monitoring, consumed-capacity estimation, and battery failsafe management through ArduPlane.

C. Electronics Layout

To secure all electronics within TUZA, custom 3D-printed mounts were designed to fully utilize the T2 Cruza's wooden mounting structure. Component placement considered center-of-gravity control, cooling, wiring accessibility, and electromagnetic interference. High-current wiring was separated from sensitive signal cables, while GPS and RF hardware were positioned away from batteries, ESCs, and power-conversion components. The final electronics layout is shown in Fig. 3.

D. Drop Mechanism and Landing Gear

Due to the limited internal volume of the Heewing T2 fuselage, TUZA uses a fully external, 3D-printed payload release assembly. The payload is secured to the assembly by a Ranger Band attached to a retaining pin. During release, a servo retracts the pin, allowing the band to disengage and the payload to separate from the aircraft before descending under a parachute.

Because TUZA lands by skidding along the fuselage, the assembly also incorporates landing gear to absorb impact and protect the release mechanism. The assembly was positioned to maintain the UAS's



Fig. 3. TUZA internal electronics layout.

center of gravity and hard-mounted to the fuselage to withstand flight and landing loads.

E. Software Design

For the mapping task, the system will generate a lawnmower-style flight pattern based on the expected flight altitude, camera field of view, and required forward and side overlap. At a given altitude, the approximate ground footprint of each image can be determined from the camera's horizontal and vertical fields of view. The spacing between adjacent flight lines is then selected to achieve the desired side overlap, while the spacing between images along each flight line is selected to achieve the desired forward overlap.

Rather than relying on a fixed image interval, the system will use the aircraft's recorded position to select imagery based on distance traveled along the mapping path. This maintains more consistent overlap when groundspeed changes because of wind, control corrections, or other flight conditions. The required along-track spacing is calculated from the

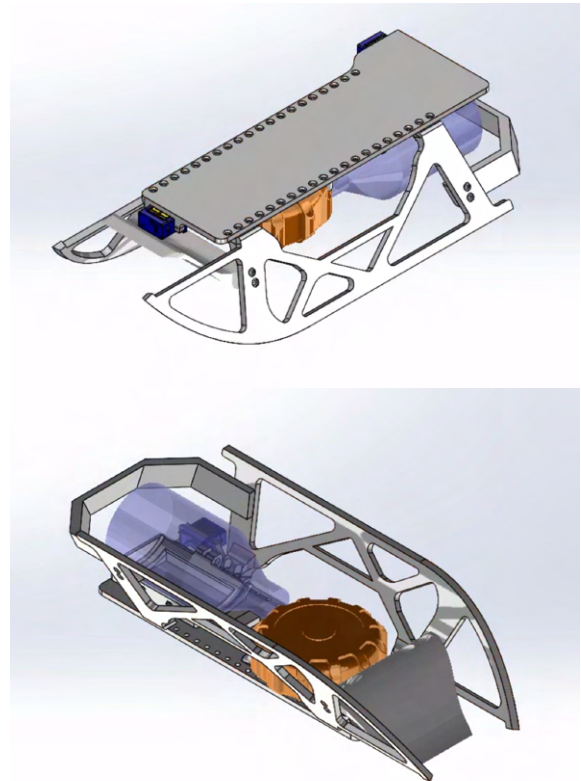


Fig. 4. Payload release assembly, CAD rendering.

camera's ground footprint and desired forward overlap and can be updated using the aircraft's actual altitude.

Imagery used for photogrammetry will be limited to the straight survey legs of the mapping pattern. Frames collected during turns will be excluded because banking, heading changes, and oblique viewing angles can reduce overlap consistency and introduce distortion. The SIYI A8 Mini's three-axis gimbal will remain pointed downward at nadir to compensate for aircraft attitude changes and reduce perspective variation between frames.

At the same time, a custom-trained YOLO26 model will be running in tandem to detect the search and delivery target without initially dropping the payload. The geotagged coordinate of the confirmed target's image frame will be saved for the delivery mission.

Before the mission, the target-localization method will be calibrated using reference objects placed at known distances and imaged from multiple altitudes. This will establish the relationship between

pixel displacement and physical ground distance. During flight, the pixel offset between the image center and the target's bounding-box center will be combined with the calibration data, aircraft position, altitude, and camera orientation to refine the estimated target coordinates [13].

After landing, frames will be extracted from the 4K recording according to the required distance spacing. Visual feature matching will associate the SD card footage with the live recording and synchronized flight-controller data, allowing GPS position, attitude, and timing information to be assigned to each frame. The resulting geotagged images will then be processed using OpenDroneMap to generate the final orthomosaic and other mapping products. TUZA will then return to the search area to execute the payload-delivery mission.

III. TESTING STRATEGY

With any engineering challenge, testing and experimentation play a vital role in validating system performance and identifying design issues. TUZA's testing ethos follows a staged approach, progressing from simulation and component-level bench testing to subsystem integration and full-flight validation. Test priorities were based on safety, technical risk, mission reliability, and available development time, allowing the team to balance iterative improvements with the repetition required to demonstrate reliable performance.

A. Power

The three parallel batteries can supply 18 A continuously, 33 A for two minutes, and 48 A for ten seconds. Battery voltage, current, and consumed capacity are recorded through the 3DR power module to verify that the system remains within these operating limits. Testing progresses from static propulsion tests to full-flight endurance tests, with particular attention given to takeoff and steep turns, which are expected to produce the highest current demand. A test is considered successful if current remains within specified limits, battery voltage does not fall below the configured failsafe threshold, and no excessive heating or abnormal voltage sag occurs.

Extended flights are then conducted to measure cruise-current draw, usable battery capacity, flight

time, and remaining charge after each lap. Multiple flights are performed under similar conditions to evaluate repeatability. These results are used to estimate the number of endurance laps TUZA can reliably complete and to determine the energy reserve available for mapping, target identification, payload delivery, and landing.

B. Communication System

Each communication link, across the different radio frequencies, was evaluated on the ground for latency, packet loss, link stability, and throughput independently before full-flight integration.

For the 2.4 GHz RC link, RCIN/RCOUT logs recorded in Mission Planner were used to determine command latency. ELRS packet rates were adjusted through the transmitter's ELRS Lua script to evaluate the trade-off between update rate, latency, and link margin, while link quality and RSSI were monitored to characterize connection health [11].

The 900 MHz telemetry link will be evaluated using MAVLink PING messages sent through MAVProxy at increasing distances to measure round-trip delay [12]. Sustained throughput will be determined by timing a known-size MAVFTP log download while monitoring Mission Planner's radio-status data for signal quality and transmission errors.

For the 5 GHz video link, end-to-end latency between the onboard Ubiquiti Bullet AC and the ground-station Rocket AC was measured by comparing timestamps from the Jetson Orin Nano live video stream with those displayed at the ground station at increasing distance. Transfer speed, packet loss, and link stability were monitored simultaneously.

C. Payload Drop Testing

To verify the payload mechanism, separate full-scale release assemblies were tested for the strobing beacon and water bottle payloads. Each assembly underwent 10 isolated release tests to evaluate payload retention, servo actuation, pin retraction, Ranger Band disengagement, and consistent separation. Additional drop tests from a known height were used to measure descent rate to facilitate selection of an appropriately sized parachute for each payload.

D. Thermal Soak Testing

Because TUZA's electronics are tightly packaged within the T2 Cruza fuselage, a thermal soak test evaluates heat buildup under direct sunlight. With the propellers removed and motors disarmed, the fully integrated UAS is powered in a mission-ready state, and the temperature of the Jetson, flight controller, Bullet AC, and other electronics is monitored throughout the test. The system must remain below component temperature limits without thermal throttling, shutdowns, communication loss, or other faults.

E. Flight Simulation Testing

ArduPilot Software-In-The-Loop (SITL) simulation was used alongside ROS2 and Gazebo to test software interaction with the flight controller, autonomous navigation, and mission execution in a safe environment. A simulated onboard camera feed was also used to test the object-detection pipeline under varying altitudes, viewing angles, and flight conditions. This use of simulation allowed extensive testing of different mission plans and guidance techniques to identify the most effective and reliable methods for completing the flight mission. After sufficient SITL simulation testing, Hardware-In-The-Loop testing was conducted to ensure proper communication and mission execution between the flight controller, companion computer, and ground control station before physical flight tests.

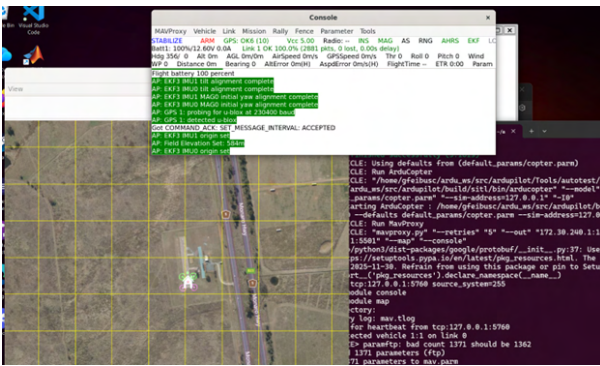


Fig. 5. ArduPilot SITL simulation.

F. Photogrammetry Testing

The photogrammetry pipeline was tested to identify OpenDroneMap settings that provide a practical balance between reconstruction speed, map

coverage, and geometric accuracy – an important consideration since imagery collected during flight must be converted into a high-quality orthomosaic without excessive processing delay.

At the time of writing, photogrammetry testing was conducted prior to obtaining a custom dataset. A public, fixed 225-image aerial dataset with characteristics similar to those predicted for the mapping task was used [1], processed using combinations of five feature-quality levels, six feature and matcher configurations, two matcher-neighbor settings, and two minimum-feature counts. All tests were run on the same computer with the same input imagery and fixed orthophoto-generation settings so that differences could be attributed primarily to the tested parameters. Recorded metrics included processing time, reconstructed area, ground sampling distance, number of reconstructed images, feature-track length, GPS error, and reprojection error. Complete averaged results are provided in the Test Results appendix.

G. Mission-Ready Flight Testing

The physical flight test followed a risk-reduction sequence that gradually increased UAS autonomy. Initial flights were conducted without the payload drop assembly and under manual control to assess characteristics of the UAS such as stability, vibration, control response, and maneuverability. AutoTune was then used to refine flight-control gains before progressing to assisted flight modes such as FBWA and autonomous navigation. Once satisfactory baseline performance was demonstrated, the payload drop assembly was installed, and flight testing was repeated to evaluate its effects on stability, center of gravity, maneuverability, and landing capability of the 3D-printed skid.

Imaging and autonomy testing began only after the UAS demonstrated reliable behavior in assisted flight modes. Mapping and object-detection pipelines were then evaluated in flight, followed by autonomous mission uploads generated from target detection.

Payload testing followed a similar staged approach: initial releases were performed manually to verify payload retention, actuation, pin retraction, and drop trajectory. After the mechanism demonstrated reliable operation, testing progressed to com-

plete autonomous delivery missions to validate waypoint generation, approach accuracy, release timing, and payload placement.

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We would also like to thank the previous members of MUAS, who helped provide a foundation for what we are building today. Their experience helped us solve problems that arose and saved the team significant time in developing solutions.

Lastly, our flight testing was conducted at our home RC field, Free State Aeromodelers – Konterra Model Airpark. Their generosity in lending their space to us was critical to the design and testing of the drone, and we are grateful to all members who share the space with us and to the staff who facilitate our presence.

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APPENDIX A

PHOTOGRAMMETRY TEST RESULTS

The appendix tables summarize the average result for each parameter value across all configurations containing that value. Bold entries identify the strongest result within each parameter group for the corresponding metric.

The tests showed a consistent tradeoff between processing speed and reconstruction completeness. Increasing feature quality generally reduced reprojection error and reconstructed more images, but substantially increased runtime. Using eight matcher neighbors produced much greater area coverage and image reconstruction than four neighbors, although it also required considerably more processing time. The FLANN-ORB configuration provided strong coverage and relatively fast processing, while FLANN-SIFT produced lower reprojection error. Increasing the minimum feature count from 3,000 to 10,000 had comparatively little effect on reconstruction quality, suggesting that the additional processing cost may not be necessary for imagery with sufficient visual features.

Typically, results would be compared by changing only a single parameter at a time while holding the rest constant, in addition to analyzing cross-parameter effects; as an example, however, Figs. 6 and 7 show two sample output orthomosaics of significantly different quality with several parameters changed simultaneously. Artifacts examined include visual area coverage, distortion of straight lines, and jagged seams. The team then selects the



Fig. 6. Sample orthomosaic: feature quality high, FLANN-ORB matcher, 8 matcher neighbors, 10,000 minimum features.



Fig. 7. Sample orthomosaic: feature quality medium, FLANN-AKAZE matcher, 4 matcher neighbors, 3,000 minimum features.

TABLE IV
AVERAGE ODM RECONSTRUCTION SIZE AND RUNTIME METRICS

Parameter	Value	Time (s)	Area (m ²)	GSD (cm/px)	Shots
feature_quality	lowest	259.35	169722.59	3.27	159.58
	low	322.10	178010.65	3.27	173.29
	medium	437.46	179718.69	3.29	175.92
	high	566.19	181728.75	3.32	180.42
	ultra	567.92	179710.83	3.33	178.33
matcher_feature	bow-hahog	479.21	181074.26	3.33	179.25
	flann-akaze	352.77	175167.21	3.28	171.65
	flann-dpsift	407.66	167109.69	3.31	155.00
	flann-hahog	496.43	181196.83	3.32	179.25
	flann-orb	343.74	182709.36	3.23	180.35
	flann-sift	503.80	177252.46	3.31	175.55
matcher_neighbors	4	341.88	125977.35	3.29	125.90
	8	519.33	228859.25	3.31	221.12
min_num_features	3000	368.24	176984.80	3.28	173.02
	10000	492.96	177851.80	3.31	174.00

TABLE V
AVERAGE ODM QUALITY AND ERROR METRICS

Parameter	Value	Track Len.	GPS Err. (m)	Reproj. Err. (px)
feature_quality	lowest	3.91	3.10	1.92
	low	3.81	2.68	1.70
	medium	3.62	2.28	1.40
	high	3.44	2.44	1.12
	ultra	3.09	2.76	0.90
matcher_feature	bow-hahog	3.53	2.73	1.26
	flann-akaze	3.42	2.85	1.42
	flann-dpsift	4.27	2.44	1.55
	flann-hahog	3.85	2.63	1.32
	flann-orb	2.92	2.55	1.77
	flann-sift	3.47	2.71	1.14
matcher_neighbors	4	3.23	3.15	1.34
	8	3.92	2.16	1.48
min_num_features	3000	3.59	2.64	1.42
	10000	3.57	2.67	1.40

configuration that results in the best balance of both qualitative and quantitative output metrics.